

NEW CASTLE, PA, USA

WMO: 720378

Lat: 41.033N

Lon: 80.417W

Elev: 1073

StdP: 14.13

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 00122

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.4	(c) 6.0	(d) -7.6	(e) 3.8	(f) 2.8	(g) -2.2	(h) 5.1	(i) 7.5	(j) 23.0	(k) 31.6	(l) 19.7	(m) 29.8	(n) 4.4	(o) 250	(p) 0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.9	(c) 89.9	(d) 74.2	(e) 87.7	(f) 73.2	(g) 84.4	(h) 71.4	(i) 76.9	(j) 85.6	(k) 75.2	(l) 83.2	(m) 73.7	(n) 81.0	(o) 7.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 74.3	(b) 133.2	(c) 80.8	(d) 72.7	(e) 126.4	(f) 78.8	(g) 71.8	(h) 122.1	(i) 77.9	(j) 40.9	(k) 85.7	(l) 39.2	(m) 83.4	(n) 37.9	(o) 80.4	(p) 82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.9	(b) 16.9	(c) 14.4	(e) -7.6	(f) 92.7	(g) 5.7	(h) 2.8	(i) -11.7	(j) 94.7	(k) -15.0	(l) 96.4	(m) -18.2	(n) 97.9	(o) -22.3	(p) 99.9		
(4) 19.9	(5) 16.9	(6) 14.4	(7) -7.4	(8) 79.3	(9) 6.6	(10) 1.6	(11) -12.2	(12) 80.4	(13) -16.0	(14) 81.3	(15) -19.7	(16) 82.2	(17) -24.5	(18) 83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	25.9	27.8	37.7	49.8	60.9	67.8	72.4	70.4	64.4	52.8	40.8	32.9
	DBStd	18.66	12.26	12.31	11.52	9.99	8.73	6.34	5.24	5.14	7.17	9.14	9.37	9.67
	HDD50	2837	751	627	414	126	13	0	0	0	0	72	301	533
	HDD65	6022	1211	1041	847	465	185	43	6	12	98	394	726	994
	CDD50	2997	5	6	33	119	350	535	695	633	432	159	26	4
	CDD65	708	0	0	0	8	58	128	237	180	80	16	1	0
	CDH74	6541	0	0	9	128	666	1171	2197	1503	737	128	2	0
	CDH80	2040	0	0	1	22	178	352	789	437	234	27	0	0
	WSAvg	5.3	6.6	6.3	6.4	6.6	5.0	4.5	3.8	3.4	3.7	5.3	5.8	6.2
	PrecAvg	36.8	2.4	2.0	2.9	3.1	3.5	3.8	4.1	3.4	3.5	2.6	3.0	2.6
(15) Precipitation	PrecMax	54.7	7.0	4.2	5.6	6.2	7.7	8.4	10.6	7.0	10.9	5.7	10.2	5.7
	PrecMin	27.6	0.6	0.2	0.9	0.7	0.9	0.7	1.2	1.3	0.6	0.3	0.7	0.7
	PrecStd	6.4	1.3	0.9	1.1	1.3	1.6	1.8	2.1	1.6	1.9	1.3	1.5	1.0
	PrecStd	6.4	1.3	0.9	1.1	1.3	1.6	1.8	2.1	1.6	1.9	1.3	1.5	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.5	65.8	74.9	82.4	88.5	90.8	93.1	90.4	90.4	84.0	70.4	62.9
		MCWB	56.1	56.6	60.1	62.4	70.9	74.1	75.6	75.2	73.7	71.9	57.7	57.2
	2%	DB	55.5	57.6	66.7	78.4	84.3	87.9	90.3	87.9	86.6	77.2	64.5	57.1
		MCWB	52.5	51.5	55.3	60.9	68.8	72.6	74.7	73.9	73.1	65.2	54.9	52.7
	5%	DB	49.1	52.3	62.6	72.7	81.6	84.0	87.8	84.4	82.0	72.8	61.1	53.6
		MCWB	45.8	48.1	52.7	58.0	66.7	71.1	74.0	72.0	70.3	63.4	53.4	50.0
	10%	DB	44.1	45.9	56.8	67.5	77.7	81.1	84.2	81.9	78.5	68.2	56.4	47.9
		MCWB	40.3	41.3	49.4	54.3	64.9	68.6	71.7	70.1	68.4	61.8	49.8	44.6
	0.4%	WB	56.7	58.4	61.8	65.4	73.8	76.4	79.7	78.2	77.1	72.9	62.0	58.6
		MCDB	59.9	63.9	72.0	77.9	83.6	85.9	88.7	86.9	87.0	81.7	67.0	61.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	53.1	53.3	58.4	62.8	71.2	74.7	77.5	75.9	74.6	68.7	57.6	54.2
		MCDB	56.0	55.9	65.9	74.2	80.9	84.1	86.6	83.5	82.4	73.4	62.0	56.5
	5%	WB	46.3	47.8	54.7	60.1	69.1	72.9	75.6	74.3	72.4	65.5	54.1	49.9
		MCDB	48.4	53.0	60.3	69.3	78.2	81.3	83.6	80.9	79.3	69.9	59.8	52.5
	10%	WB	40.1	41.3	50.2	57.2	66.9	71.0	73.9	72.6	70.2	62.6	50.5	44.4
		MCDB	43.8	45.9	55.3	65.6	74.7	77.9	80.9	78.9	76.0	67.3	55.6	47.7
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	16.6	18.7	22.1	23.1	21.5	21.9	21.5	21.4	19.4	17.2	13.6
		MCDBR	19.8	23.7	25.5	30.0	27.5	25.0	24.6	23.9	25.7	26.1	24.6	20.4
		MCWBR	17.4	19.1	16.6	16.9	13.6	12.1	11.0	11.3	12.8	14.5	16.3	17.2
	5% WB	MCDBR	18.5	22.8	21.9	25.2	23.6	21.2	21.0	19.6	21.9	20.8	20.8	19.2
		MCWBR	17.5	19.4	16.0	16.2	12.3	11.4	10.4	9.9	11.6	13.3	16.0	18.4
			17.5	19.4	16.0	16.2	12.3	11.4	10.4	9.9	11.6	13.3	16.0	18.4
(40) Clear-Sky Solar Irradiance	taub		0.301	0.330	0.355	0.383	0.421	0.449	0.451	0.442	0.386	0.361	0.318	0.301
	taud		2.439	2.339	2.350	2.310	2.218	2.177	2.190	2.207	2.340	2.406	2.508	2.490
	Ebn at Noon		272	278	283	282	272	264	262	261	269	263	264	263
	Edh at Noon		25	32	36	40	44	46	45	43	36	30	23	22
			25	32	36	40	44	46	45	43	36	30	23	22
(44) All-Sky Solar Radiation	RadAvg		455	734	1036	1411	1667	1813	1829	1643	1330	866	573	371
	RadStd		43	93	104	145	146	152	137	91	132	84	80	37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	+101

Nomenclature: See separate page